EXPIREMENT 3 8 BIT MULTIPLCICATION

AIM:

TO write an assemby launguage program to impilement 8 bit multiplication using 8085 processer.

ALGORITHM:

1. Start the program by loading a register pair with the address of memory location.

2. Move the data to a register

3. Get the second data and load it into the accumulator.

4. Add the two register contents.

5. increment the value of the carry.

6. check whether the repeated addition is over

7. Store the value of the product.

8. Halt.

RESULT:

Thus the program was executed successfully using 8085 processor simulator.

PROGRAM:

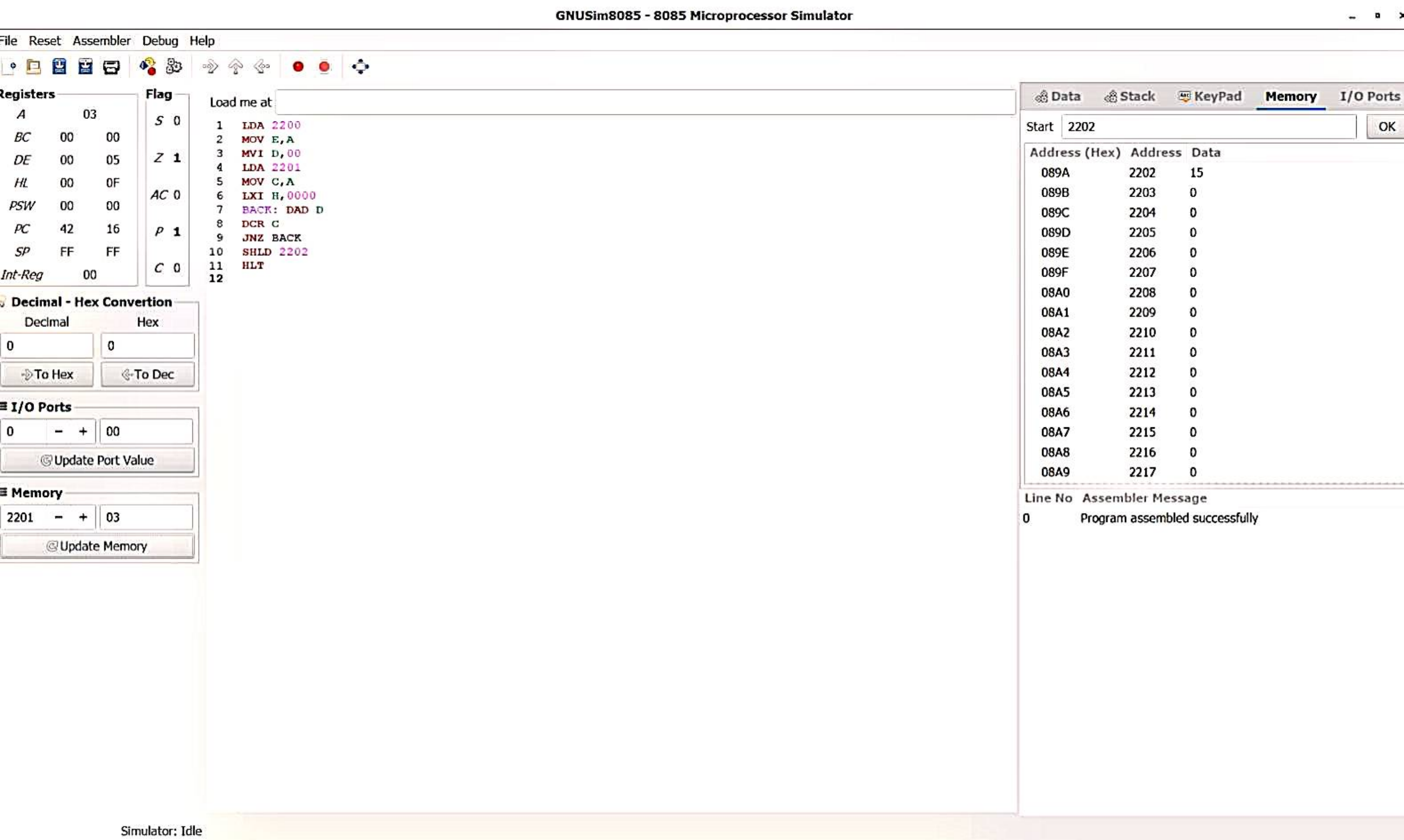
MNEMONICS	EXPLANATION
LDA 2200	Load the accumulator with the first number
MOV E A	Move the data from accumulator to E
MVI D, 00	Move the immidiate value 00 unto.
LDA 2201	Load the accumulator address.
MOV C,A	Move the data from the accumulator.
LXI H0000	Load the accumulator
BACK :DAD	Back : Label for Loop.
DCR C	Decrement register E By 1.
JNZ BACK	in register E, is not 0
SHLD 2202	Store the value on the HL
HLT	HALT

INPUT:

ADDRESS	DATA
2200	4
2201	2

OUTPUT:

ADDRESS	DATA
2202	8



GNUSim8085 - 8085 Microprocessor Simulator
File Reset Assembler Debug Help
Registers
A 03
BC 00 00
DE 00 05
HL 00 0F
PSW 00 00
PC 42 16
SP FF FF
Int-Reg 00
Flag
S 0
Z 1
AC 0
P 1
C 0
Load me at
LDA 2200
MOV E,A
MVI D,00
LDA 2201
MOV C,A
LXI H,0000
BACK: DAD D
DCR C
JNZ BACK
SHLD 2202
HLT
Decimal - Hex Convertion
Decimal Hex
0 0
To Hex To Dec
I/O Ports
0 − + 00
Update Port Value
Memory
2201 − + 03
Update Memory
Data Stack KeyPad Memory I/O Ports
Start 2202 OK
Address (Hex) Address Data
089A 2202 15
089B 2203 0
089C 2204 0
089D 2205 0
089E 2206 0
089F 2207 0
08A0 2208 0
08A1 2209 0
08A2 2210 0
08A3 2211 0
08A4 2212 0
08A5 2213 0
08A6 2214 0
08A7 2215 0
08A8 2216 0
08A9 2217 0
Line No Assembler Message
0 Program assembled successfully
Simulator: Idle